

Ali Ghasemzadeh

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Research Interests

Applied: Medical AI, Computer Vision, LLMs

Theory: Estimation Theory, Convex & Semidefinite Optimization

Education

Sharif University of Technology (SUT), Tehran, Iran

2022 – Expected Feb 2027

B.Sc. in Computer Engineering | **Minor: Mathematics**

GPA: 18.83/20

National Organization for Development of Exceptional Talents

2019 – 2022

High School Diploma in Mathematics and Physics

GPA: 19.5/20

Research Internship

The University of Hong Kong (HKU), Hong Kong

Jul 2026 – Present

Research Intern, Department of Data and Systems Engineering

Supervisor: Prof. Yong-Hong Kuo [\[link\]](#)

Developing an LLM-based framework for constructing domain-specific knowledge graphs from heterogeneous data sources, with applications to supply chain, travel, and transportation.

Publications & Manuscripts

PriLoRA: Prior-Conditioned Low-Rank Adapters for Medical Vision Models, Amirhossein Kazerouni [\[link\]](#), Sanaz Karimi jafarbigloo [\[link\]](#), Ali Aghayari [\[link\]](#), **Ali Ghasemzadeh**, Reza Azad [\[link\]](#), Wojciech Samek [\[link\]](#), Dorit Merhof [\[link\]](#), Michael Brudno [\[link\]](#), Babak Taati [\[link\]](#). **Accepted at MICCAI 2026**. Co-developed the prior-conditioned low-rank adaptation framework and implemented its evaluation on medical image segmentation and classification tasks.

Toward Accurate RUL and SOH Estimation Using Reinforced Graph-Based PINNs Enhanced with Dynamic Weights, Mohamadreza Akbari Pour, **Ali Ghasemzadeh**, MohamadAli Bijarchi [\[link\]](#), Mohammad Behshad Shafii [\[link\]](#). **Accepted at Advanced Engineering Informatics**. Proposed the reinforced graph-based PINN methodology, implemented the pipeline, and contributed to the manuscript for battery RUL and SOH estimation. [arXiv](#)

A Review of Reward Engineering for Reinforcement Learning Guided LLM Metaheuristic Generation, **Ali Ghasemzadeh**, Armin Khosravi, Mohamad Mirzadi, Mohamadreza Akbari Pour, Seyedali Mirjalili [\[link\]](#). **Minor revision at Archives of Computational Methods in Engineering**. Co-first author. Co-led the survey design, developed the taxonomy and future-directions, and coordinated the synthesis of reward-engineering strategies. [SSRN](#)

Spectral Palette: Learning Diverse Spectral Bases for Implicit Neural Representations, Amirhossein Kazerouni [\[link\]](#), **Ali Ghasemzadeh**, Armin Khosravi, Donya Jafari, Erfan Babakhani, Ali Diba [\[link\]](#), Yong-Hong Kuo [\[link\]](#), Michael Brudno [\[link\]](#), Babak Taati [\[link\]](#). **Submitted to AAAI**. Co-developed the core idea, implemented the full experimental pipeline, and authored the experimental analysis for learning diverse spectral bases in INRs.

GNBG-C: A Unified and Configurable Benchmark Generator for Constrained Continuous Optimization, **Ali Ghasemzadeh**, Armin Khosravi, Erfan Babakhani, Pariya Ghasemzadeh, Yong-Hong Kuo [\[link\]](#), Seyedali Mirjalili [\[link\]](#), Amir H. Gandomi [\[link\]](#). **Submitted to European Journal of Operational Research**. Proposed and implemented the benchmark generator, including controllable feasible-region properties and known reference optima, and contributed to the manuscript.

Learning Under Joint Domain Evolution: Progressive Multi-Source Domain Adaptation, Ali Sadeghian, **Ali Ghasemzadeh**, Seyedreza Tavakoli, Amir Naghdi, Aria Mokhtari, Mahdi Ghiyasi, Babak Khalaj [\[link\]](#), Farzan Farnia [\[link\]](#), Soheil Kolouri [\[link\]](#), Ali Diba [\[link\]](#). **Submitted to AAAI**. Co-first author. Co-led the problem formulation, developed the progressive multi-source adaptation method, implemented the experiments, and contributed to the manuscript.

Survey on Diffusion Models for Combinatorial Optimization, Armin Khosravi, **Ali Ghasemzadeh**, Erfan Babakhani, Amirhossein Kazerouni [\[link\]](#), Yong-Hong Kuo [\[link\]](#), Philip Torr [\[link\]](#), Seyedali Mirjalili [\[link\]](#). **Submitted to Intelligent Systems with Applications**. Co-first author. Co-led the literature synthesis and developed the taxonomy and comparative analysis of diffusion-based methods for combinatorial optimization. [SSRN](#)

Learning from Many Minds: A Survey of Multi-Rater Medical Image Segmentation under Annotation Ambiguity, Erfan Babakhani, Armin Khosravi, **Ali Ghasemzadeh**, Pariya Ghasemzadeh, Sanaz Karimi jafarbigloo [\[link\]](#), Reza Azad [\[link\]](#), Ulas Bagci [\[link\]](#), Dorit Merhof [\[link\]](#). Co-first author. Co-led the literature review and taxonomy design for multi-rater medical image segmentation under annotation ambiguity, and co-wrote the manuscript.

Reference-Based Medical Image Segmentation Survey, Erfan Babakhani, **Ali Ghasemzadeh**, Armin Khosravi, Pariya Ghasemzadeh, Amirhossein Kazerouni ^[link], Yong-Hong Kuo ^[link], Seyedali Mirjalili ^[link]. *Co-first author*. Co-led the literature curation and analysis, identified methodological strengths and limitations, and developed the taxonomy and comparative framework.

Research Experience

Multi-Armed Bandit for Adaptive Metaheuristic Selection Apr 2026 – Present

Advisors: Prof. Seyedali Mirjalili ^[link] (Torrens University Australia), Prof. Amir H. Gandomi ^[link] (UTS)

Proposed and implemented a bandit framework that treats evolutionary algorithms as arms and adaptively selects among them during optimization across heterogeneous tasks.

Excess Estimation Error May 2026 – Present

Advisors: Dr. Alex Dytso ^[link] (Qualcomm / Princeton University), Prof. Luca Barletta ^[link] (Politecnico di Milano)

Studying excess estimation error as a probability-of-error criterion; derived functional properties, established basic bounds, and characterized the high-SNR regime.

Estimation Theory: Inverse CDF Differentiation Mar 2026 – May 2026

Advisor: Dr. Alex Dytso ^[link] (Qualcomm / Princeton University)

Derived differentiation results for inverse cumulative distribution functions and analyzed implications for statistical estimation and information-theoretic quantities.

Structural Properties of Semidefinite Programs Nov 2025 – Mar 2026

Advisor: Prof. Peyman Mohajerin Esfahani ^[link] (University of Toronto)

Studied inconvexity, pseudoconvexity, and hidden convexity in semidefinite programming, focusing on tractable convex reformulations of nonconvex problems.

Adaptive Batched Adversarial Linear Bandit Jun 2025 – Dec 2025

Advisors: Dr. Hossein Esfandiari ^[link] (Google), Dr. Samira Hossein Ghorban ^[link] (IPM)

Contributed the key proof idea for a lower bound in the batched setting and worked on regret analysis under non-adaptive feedback.

Strategic Producer Modeling and Market Mechanism Design Apr 2025 – Jul 2025

Advisor: Prof. Alireza Fallah ^[link] (Rice University)

Developed a game-theoretic framework for strategic producers in online markets and analyzed equilibrium behavior, pricing strategies, and learning dynamics.

Honors & Awards

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| • Ranked 1st in Estimation Theory, a graduate-level EE course taken as an undergraduate | 2025 |
| • Ranked 2nd among 60 EE and CE students in Convex Optimization | 2025 |
| • Top 0.03% in the Iranian National University Entrance Exam (Konkour) | 2022 |

Research Software

GNBG-C: Open-source configurable benchmark generator for constrained continuous optimization with controllable feasible-region properties and known reference optima. [\[GitHub\]](#)

PriLoRA: Official implementation of Prior-Conditioned Low-Rank Adapters for Medical Vision Models (MICCAI 2026). [\[GitHub\]](#)

Advanced Coursework

AI & ML: Artificial Intelligence, Machine Learning, Reinforcement Learning, Generative Models

Math & Optimization: Convex Optimization, Estimation Theory, Stochastic Processes, Mathematical Analysis, Game Theory, Linear Algebra, Probability & Statistics, Numerical Computation

Teaching Assistant Experience (Designed theoretical and practical assignments)

Machine Learning Theory: Dr. Najafi (Fall 2025)

Machine Learning: Dr. Motahari (Spring 2025)

Probability & Statistics: Dr. Najafi (Spring & Fall 2024)

Linear Algebra: Dr. Rabiee & Dr. Ramezani (Fall 2024)

Automata Theory: Dr. Dolati (Fall 2024)

Fundamentals of Programming: Dr. Fazli (Fall 2023)

Technical Skills

ML PyTorch, TensorFlow, Hugging Face, PEFT, CVXPY, NumPy, SciPy, Pandas, scikit-learn

Programming Python, C/C++, MATLAB

Tools Git, Linux, Docker, L^AT_EX

Languages Persian (Native), English (Professional)